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NỀN TẢNG LẬP TRÌNH CHO PHÂN TÍCH VÀ TRỰC QUAN DỮ LIỆU

Programming Foundations for Data Analysis and Visualization

ASSIGNMENT REPORT · PART 3 OF 3 · TIME-SERIES DATA

Forecasting Hanoi's Daily Temperature

Twenty-one years of weather, winter cold surges, and how far ahead a forecast stays useful

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[Project page](#) · [Notebook on GitHub](#) · Video: to be published

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1 Problem and objective

Hanoi has a real winter: cold surges of the north-east monsoon can cut the temperature by 5–10 °C within a day, and those are exactly the days when a forecast matters most for health, energy use and agriculture.

Objective. Forecast **tomorrow's daily mean temperature in Hanoi** from today's and past weather, then measure how quickly forecast skill decays when predicting up to seven days ahead.

How success is measured. Mean absolute error (MAE, °C) on a chronologically held-out test period, and *skill* relative to simple baselines — above all *persistence* (“tomorrow = today”), which is hard to beat for one-day forecasts.

2 Dataset

The data was downloaded by the group from the **Open-Meteo Historical Weather API** [1] (free, no API key, CC BY 4.0), which serves the **ERA5 reanalysis** [2]: a physically consistent reconstruction of past weather on a 9–25 km grid, combining a weather model with millions of observations. The exact request is included in the notebook (`fetch_open_meteo`); *Run all* uses the saved CSV so results are reproducible.

- Location: central Hanoi (21.03 °N, 105.85 °E; nearest grid cell 21.05 °N, 105.90 °E, elevation 19 m).
- Period: 1 January 2005 – 26 September 2026, daily, local time (GMT+7): 7,939 consecutive days, 14 variables.

Table 1. Variables. The target is the daily mean temperature.

Variable	Unit	Meaning
temperature_2m_mean / _max / _min	°C	daily mean (target), maximum, minimum at 2 m
apparent_temperature_mean	°C	“feels-like” temperature
precipitation_sum, precipitation_hours	mm, h	daily rain and rainy hours
relative_humidity_2m_mean, dew_point_2m_mean	%, °C	moisture
pressure_msl_mean	hPa	mean sea-level pressure
cloud_cover_mean	%	cloud cover
wind_speed_10m_max, wind_direction_10m_dominant	km/h, °	maximum wind speed, dominant direction
shortwave_radiation_sum, sunshine_duration	MJ/m ² , s	solar energy, sunshine

Table 2. General requirements (section 4.1 and 5) and how this dataset meets them.

Requirement	This dataset	Status
Enough samples	7,939 days	Met
Numerical columns (scaling)	14 weather variables plus engineered lags	Met
Categorical columns (encoding)	derived: season and 8-sector wind direction (one-hot), circular encodings	Met
Missing values handled	none in the source (verified); gaps simulated to compare imputation methods (Section 4.1)	Met
Time-aware split	train / validation / test by date; time-series cross-validation	Met

3 Exploratory data analysis

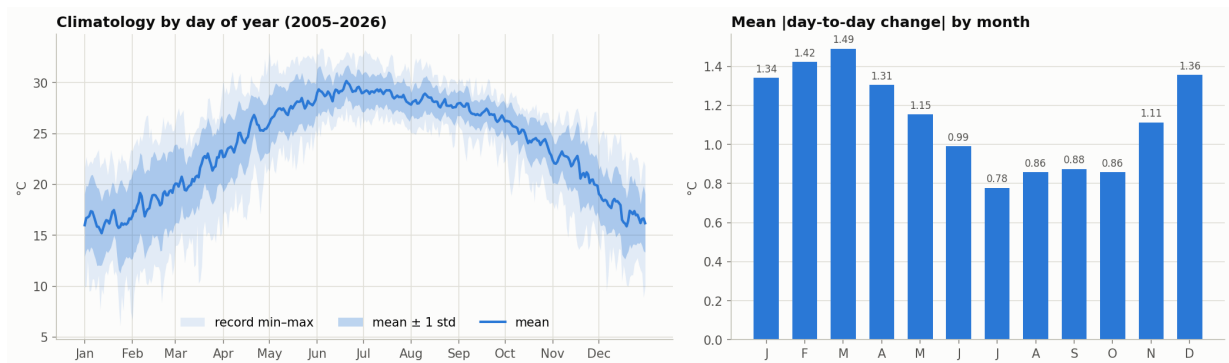


Figure 1. Left: climatology by day of year (mean, ± 1 standard deviation, record range). Right: mean absolute day-to-day change by month.

Question. How different are the seasons — and when is the weather hardest to predict?

Finding. Summer (June–August) averages 28.7 °C and winter (December–February) 17.4 °C, a swing of about 11 °C. Winter is also far more *variable*: the standard deviation within January is 3.40 °C versus 1.62 °C in July, and the average day-to-day change peaks in February–March (1.42–1.49 °C) against 0.78 °C in July. Forecasting should therefore be hardest in winter and spring.

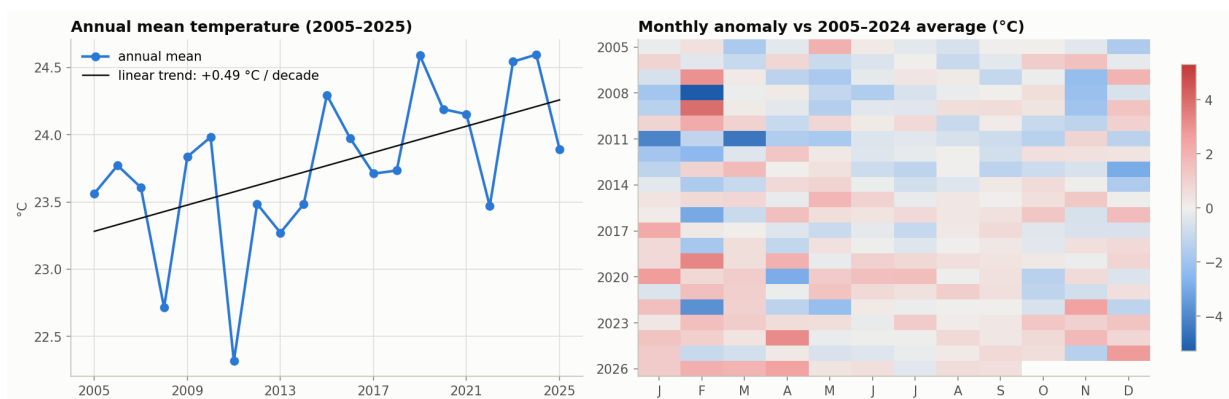
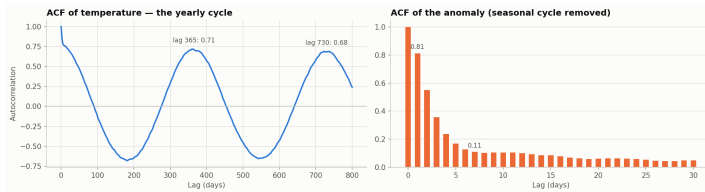


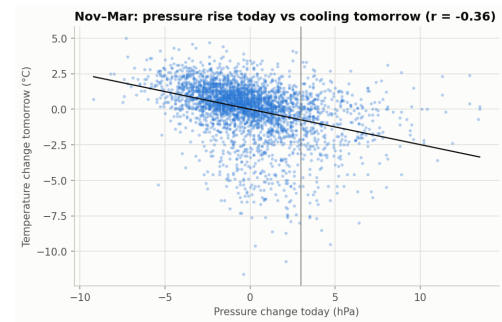
Figure 2. Left: annual mean temperature with a linear trend. Right: monthly anomaly versus the 2005–2024 average (blue = colder, red = warmer).

Question. Is Hanoi getting warmer?

Finding. The annual mean rises by +0.49 °C per decade over 2005–2025; 2021–25 averaged 24.13 °C against 23.50 °C in 2005–09 (+0.63 °C). Large cold anomalies cluster in the early years (February 2008, January–February 2011) and recent years are mostly warm. Twenty-one years is short and year-to-year swings are large (22.3 °C in 2011 vs 24.6 °C in 2024), so the slope is uncertain — but it matters for modelling: a climatology learned on 2005–2021 will be slightly too cold for 2024–2026 (Section 6).



(a) Autocorrelation of temperature and of its anomaly.



(b) Pressure change today vs temperature change tomorrow (Nov–Mar).

Figure 3. Memory of the series and the cold-surge signal.

Question. How much does today tell us about tomorrow, and can we see a cold surge coming?

Finding. The raw series is dominated by the yearly cycle (autocorrelation 0.71 at lag 365). Once the season is removed, the anomaly still has strong day-to-day memory (**0.81 at lag 1**) that fades to **0.11 by lag 7**: a warm or cold spell “remembers” itself for only a few days. In the cold season a pressure rise today warns of cooling tomorrow ($r = -0.36$): on the 404 days when pressure rose by more than 3 hPa, the next day changed by $-1.03\text{ }^{\circ}\text{C}$ on average (vs $+0.15\text{ }^{\circ}\text{C}$ otherwise) and a drop of more than $2\text{ }^{\circ}\text{C}$ was **2.7× as likely** (26.5% vs 9.6%).

4 Data preparation

4.1 Continuity and missing values

A time series must be checked for missing *dates*, not only missing cells, because a skipped day silently shifts every lag feature. The file has 7,939 rows for 7,939 calendar days, no duplicated dates and 0 missing cells. Real station data is rarely this clean, so we used the complete series as ground truth for an experiment: hide random blocks of 1–30 consecutive days (about 5% of the series, 5 repetitions) and reconstruct them with four methods.

Table 3. Imputation error (MAE in °C on the hidden days; best per gap length in bold).

Gap length	Forward fill	Linear interp.	Climatology	Seasonal interp.
1 day	1.16	0.71	1.97	0.70
3 days	1.76	1.14	1.89	1.13
7 days	2.21	1.69	2.00	1.63
14 days	2.45	1.95	1.86	1.93
30 days	2.80	2.22	1.87	2.15

Question. If a station lost data for a few days, how should the gap be filled — and does the answer depend on its length?

Finding. For short gaps, **seasonal interpolation** (interpolating the anomaly, then adding the climatology back) is best: 0.70, 1.13 and 1.63 °C for 1, 3 and 7 days, slightly ahead of plain linear interpolation (0.71, 1.14, 1.69). Forward fill is worst at every length. For gaps of **14 days or more, the seasonal normal alone wins** (1.86 °C), because the anomaly’s memory (about a week) has run out. Rule: interpolate the anomaly up to about a week, fall back to climatology for longer outages.

4.2 Outliers

Temperature outliers must be judged against the season (12 °C is normal in January but extreme in July), so each day’s anomaly is divided by the standard deviation for that day of the year and $|z| > 3$ is flagged. Only 9 days qualify, all cold (March 2011, early October 2011, September 2013): real cold spells, exactly what a forecast must handle, so they are **kept**. The record 6.1 °C of 24 January 2016 is not even flagged because January is so variable. Precipitation has 1,001 IQR outliers and skewness 5.2 (maximum 227 mm on 21 June 2010); heavy rain is also real, so it is kept and fed to the models as $\log_{1p}(\text{rain})$ (skewness 0.95).

4.3 Features, encoding and scaling

For a forecast issued at the end of day t for day $t + h$, every feature must be known on day t . The class `WeatherFeatureBuilder` builds them; its climatology is learned from the **training years only**.

Table 4. Features (30 in total) and their encoding.

Group	Features	Encoding / scaling
Recent temperature	today; lags 1, 2, 3, 7, 14 days; rolling means over 3, 7, 30 days; today's max, min and range	standardised
Anomaly	today – climatology (smoothed day-of-year mean)	standardised
Weather today	humidity, dew point, pressure, pressure change over 1 and 3 days, cloud cover, wind speed, rain, radiation, sunshine	log1p for rain; standardised
Wind direction	sine and cosine of the direction	circular (359° is next to 1°)
Calendar of target day	day of year; climatology of the target day	sine/cosine (cyclical)
Categorical	season of target day (DJF, MAM, JJA, SON); 8-sector wind direction (N, NE, ..., NW)	one-hot

The first 30 days (rolling-window warm-up) and the last day (no target) are dropped. Scaling uses `StandardScaler` inside the pipeline; it is required by Ridge regression and harmless for tree models.

4.4 Time-based split

A random split would let the model learn from days *after* the ones it is tested on. We split by the date being forecast: **train** 2005–2021 (6,179 days), **validation** 2022–2023 (730 days) and **test** 1 January 2024 – 26 September 2026 (1,000 days). Hyperparameters are tuned with `TimeSeriesSplit` (5 expanding windows inside the training years), which always trains on the past and validates on the following period.

5 Modelling

Baselines. *Persistence* (tomorrow = today); *climatology* (the average for that calendar day); and *damped anomaly persistence* (climatology + $\varphi \times$ today's anomaly, with $\varphi = 0.81$, the lag-1 autocorrelation of anomalies in the training years).

Machine-learning models. Ridge regression, Random Forest and HistGradientBoosting (HGB), all in the same pipeline. HGB is trained either on the temperature itself or on the **change** $\Delta T = T(t+1) - T(t)$, adding today's value back afterwards (`DeltaRegressor` class). Trees cannot extrapolate beyond the range seen in training, and a warming climate makes that a real risk; predicting the change avoids it.

Tuning. A random search over 20 HGB configurations (scikit-learn defaults included) with time-series cross-validation preferred a more regularised model (CV MAE 0.928 vs 0.944 for the defaults), but it did *worse* on 2022–2023 (0.922 vs 0.906). A 0.015 °C gain on older folds is within year-to-year noise, so the final model is the one chosen on validation: HGB on the change, with default hyperparameters.

Table 5. One-day-ahead forecasts: validation (2022–2023) and test (2024 – September 2026) scores. Skill = $1 - \text{MAE} / \text{MAE of persistence (test)}$.

Model	Val MAE	Test MAE	Test RMSE	Test R ²	Skill
Climatology (seasonal normal)	2.014	1.990	2.538	0.722	−84.2%
Persistence (tomorrow = today)	1.105	1.081	1.504	0.902	0.0%
Damped anomaly persistence	1.092	1.075	1.434	0.911	0.5%
Ridge regression	0.974	0.970	1.283	0.929	10.2%
Random Forest	0.921	0.935	1.245	0.933	13.4%
HGB, temperature as target	0.933	0.932	1.231	0.934	13.8%
HGB, change as target, tuned	0.922	0.919	1.222	0.935	15.0%
HGB, change as target (final)	0.906	0.916	1.226	0.935	15.3%

Question. Do machine-learning models beat the classical baselines, and which framing works best?

Finding. Every model beats every baseline. HGB predicting the change is best on validation (0.906 °C, 18% below persistence at 1.105) and beats HGB predicting the level (0.933). Ridge regression is weakest (0.974): the useful signals interact non-linearly — a northerly wind matters mostly in winter. Damped anomaly persistence barely improves on persistence (1.092) and climatology alone is poor (2.014 °C): for *tomorrow*, today's weather matters far more than the calendar.

6 Results on the test set

On the 1,000 unseen days the final model reaches **MAE 0.916 °C, RMSE 1.226 °C and R^2 0.935**, a **15.3% lower error than persistence** (1.081 °C); the ranking of all models matches validation (Table 5).

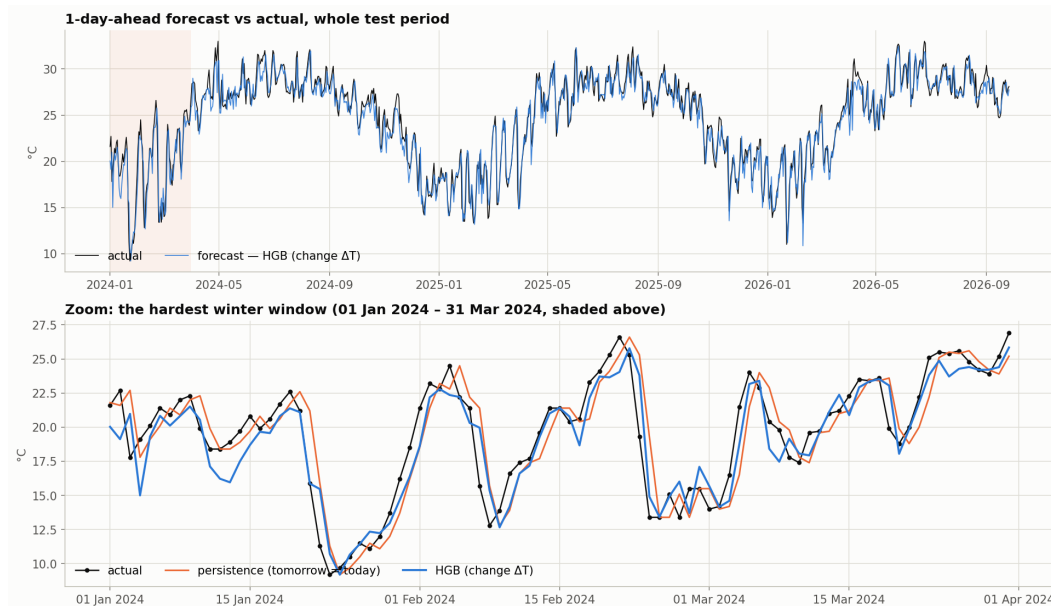


Figure 4. Forecast versus actual over the whole test period (top) and the hardest winter window (bottom), with persistence for comparison.

Table 6. Test MAE by month (bold = final model better than persistence).

Test MAE (°C)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Persistence	1.36	1.61	1.24	1.08	1.08	0.99	0.75	0.87	0.86	0.96	1.01	1.12
Final model	1.05	1.16	0.82	1.10	0.91	0.87	0.69	0.84	0.72	0.90	1.12	0.88

Table 7. The 8 largest test errors (°C; pressure change on the forecast day in hPa).

Date	Actual	Forecast	Error	1-day change	Pressure change
29 March 2025	17.0	23.7	+6.7	−8.5	+4.2
8 February 2026	15.0	20.7	+5.7	−7.0	+2.2
18 November 2025	17.8	22.7	+4.9	−4.9	−0.1
23 February 2024	19.3	23.8	+4.5	−6.0	+3.7
26 January 2025	14.9	19.4	+4.5	−6.4	+1.6
29 April 2024	31.7	27.4	−4.3	+1.0	+2.3
7 February 2024	15.7	20.0	+4.3	−5.7	+0.8
22 January 2024	11.3	15.5	+4.2	−4.6	+3.5

Question. When does the model fail?

Finding. The biggest gains over persistence come in winter and spring — February 1.16 vs 1.61, March 0.82 vs 1.24, December 0.88 vs 1.12 °C — while the model is slightly worse in the transition months (April 1.10 vs 1.08, November 1.12 vs 1.01). **7 of the 8 worst forecasts are cold surges:** the temperature fell 4.6–8.5 °C in one day and the forecast was 4.2–6.7 °C too warm, usually after a pressure rise that the model reacted to only partly. On average the residual is -0.27 °C (standard deviation 1.20): forecasts are slightly too cold, consistent with the warming of Figure 2.

7 Extra experiments

7.1 How far ahead is the model useful?

The features are rebuilt and the final configuration retrained for horizons of 1 to 7 days; all methods are scored on the same test period.

Table 8. Test MAE (°C) by forecast horizon (best per horizon in bold).

Test MAE (°C)	1 d	2 d	3 d	4 d	5 d	6 d	7 d
Persistence (tomorrow = today)	1.08	1.74	2.16	2.44	2.56	2.60	2.61
Climatology (seasonal normal)	1.99	1.99	1.99	1.99	1.99	1.99	1.99
Damped anomaly persistence	1.07	1.60	1.86	1.97	2.00	1.99	1.99
Final model	0.92	1.49	1.79	1.95	1.99	2.00	1.99

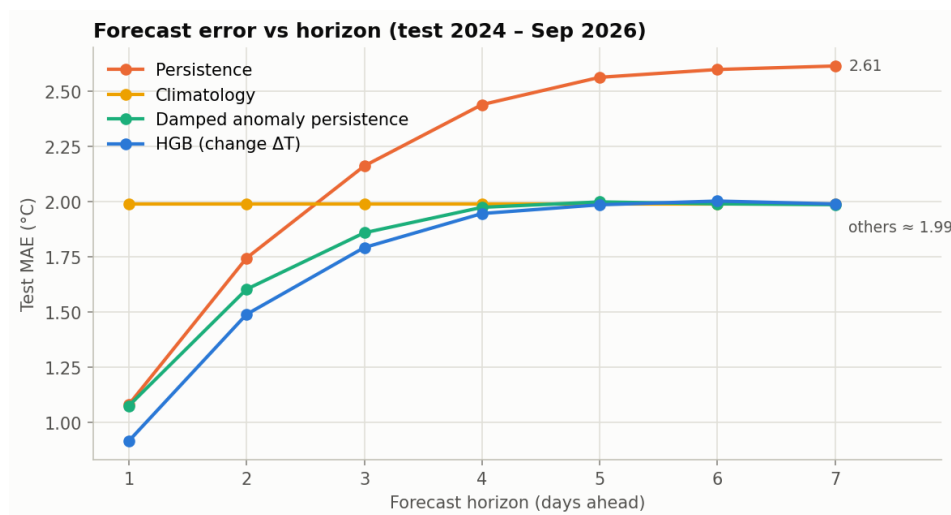


Figure 5. Forecast error versus horizon on the test period.

Question. At what horizon does the model stop beating simple baselines?

Finding. The model beats every baseline for **1 to 4 days ahead** (MAE 0.916, 1.489, 1.792, 1.946 °C vs 1.075, 1.603, 1.858, 1.974 for the best baseline). From day 5 all methods converge on climatology (about 1.990 °C): the anomaly memory has faded, exactly as the autocorrelation of Figure 3 predicted. Persistence becomes worse than the seasonal average from day 3 (2.161 °C).

7.2 What drives the predicted change?

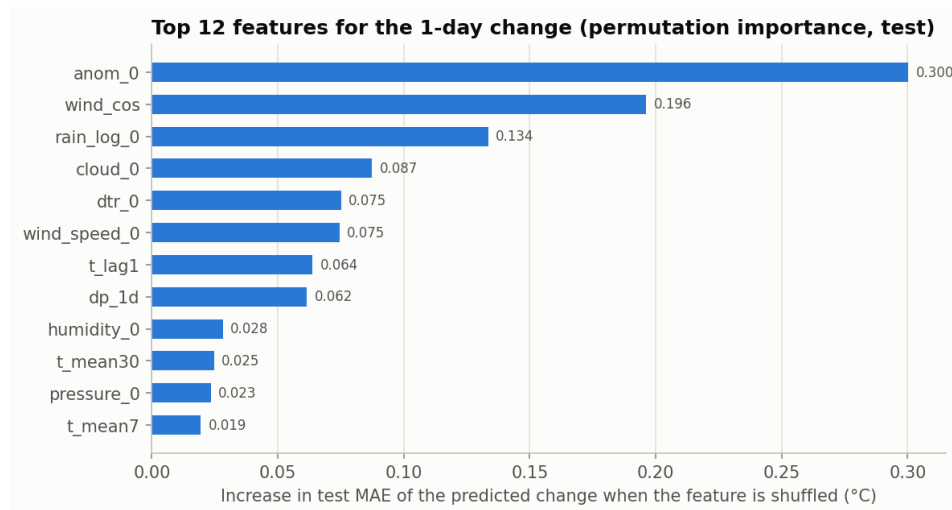


Figure 6. Permutation importance of each feature for the predicted one-day change (test set).

Question. Which signals does the model use to predict tomorrow's change?

Finding. Today's **anomaly** (0.300) comes first — an unusually warm day tends to cool back towards normal and vice versa. Next is the **north–south component of the wind** (0.196): a northerly wind brings cold air, the signature of a cold surge. Rain (0.134) and cloud cover (0.087) follow, as they suppress daytime heating. The one-day pressure change is used (0.062) but ranks lower than Figure 3 suggests, because the wind direction carries much of the same information.

8 Conclusions

1. **Machine learning beats the classical baselines, modestly:** 15.3% lower test MAE than persistence (0.916 vs 1.081 °C). Framing the target as the *change* helped more than hyperparameter tuning.
2. **The hard days are cold surges.** Winter and spring are where the model gains most and where its worst misses happen; the physical signals — northerly wind, rising pressure, to-day's anomaly — are what it learned to use.
3. **Skill fades after about 4 days**, when every method converges on the seasonal average, matching the roughly one-week memory of temperature anomalies.
4. **Data preparation lessons:** split by time, never randomly; fit the climatology on training years only; fill short gaps by interpolating the anomaly and long gaps with climatology.
5. **Hanoi is warming** by about +0.49 °C per decade in this record — enough to leave a −0.27 °C bias in forecasts trained on earlier years.

Limitations. ERA5 is a smoothed grid-cell value rather than a thermometer in the city; real station data would be noisier. The forecast uses the complete weather of day t , slightly more than an evening forecast would have. Only local variables are used, while cold surges start hundreds of kilometres to the north.

Future work. Add upstream features (pressure and temperature in southern China a day earlier) to see cold surges coming; use hourly data for 24-hour forecasts; produce probabilistic forecasts (quantile regression) to warn of large drops; compare with archived forecasts from a physics-based weather model; retrain regularly to follow the warming trend.

References

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- [5] Group G3. *Project page (rendered notebook, report and video)*. https://trungly-hcmut.github.io/C05177_DataVisualization_Assignment/timeseries/ — video: to be published